



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

**School of Computer Science
and Engineering**

SC2104/CE3002: Sensors, Interfacing and Digital Control

Topic 7 – Motors and Encoders Operations

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DC Motor

A Direct Current (DC) Motor is a rotating electrical device that convert electrical energy to mechanical energy



DC motor
with Gearbox

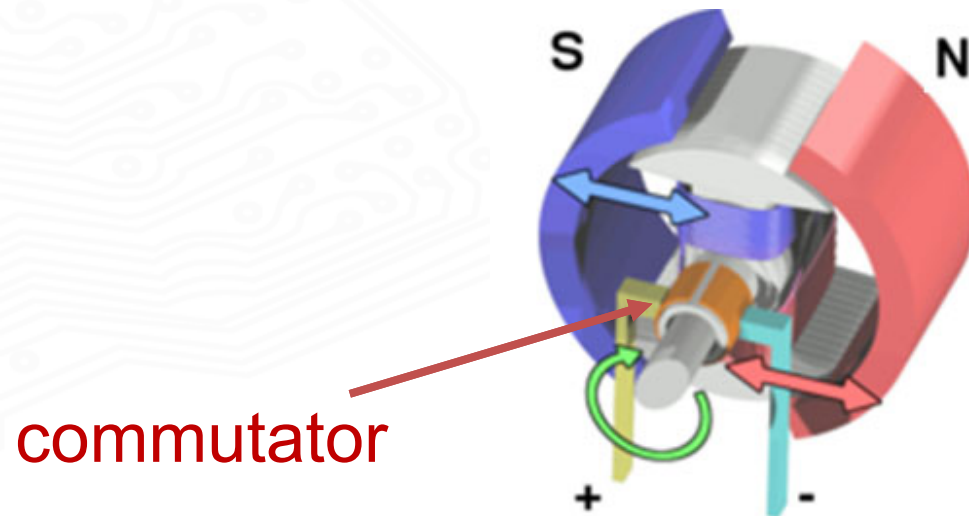
When a DC voltage is applied to its terminal

- current flows through a coil of wire (i.e., inductor) inside the motor produces a magnetic field
- create a rotary motion

DC Motor Internal

Inside the DC motor is the rotor, an iron shaft wrapped in coils of wire

- around the shaft is the stator consists of two fixed, North pole and South pole magnets



Source: Wikipedia

As the rotor shaft is rotating

- current continuous to flow to the coil through a ring commutator

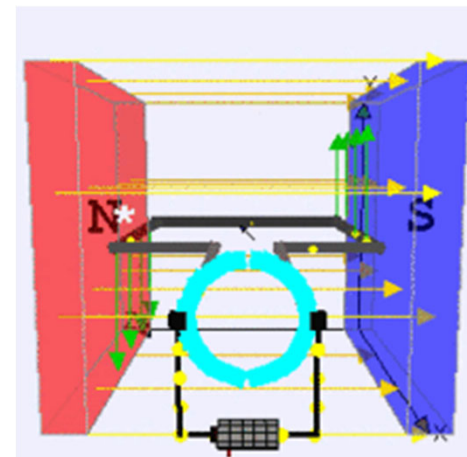
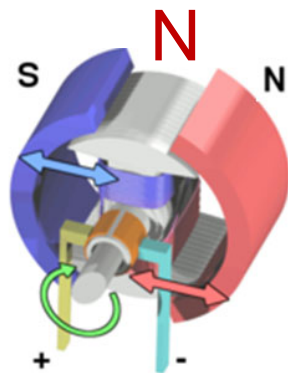
DC Motor Operation Principle

When the DC current flows through the coil

- magnetic field is generated in the coil
 - polarity depends on the direction of the current flow
- the coil is hence attracted/repulsed by the fixed magnetic
 - create the rotating motion

The coil is arranged in a way such that

- its polarity will continuously switch between North and South poles as it rotates between the two magnets
- hence maintaining the rotation motion



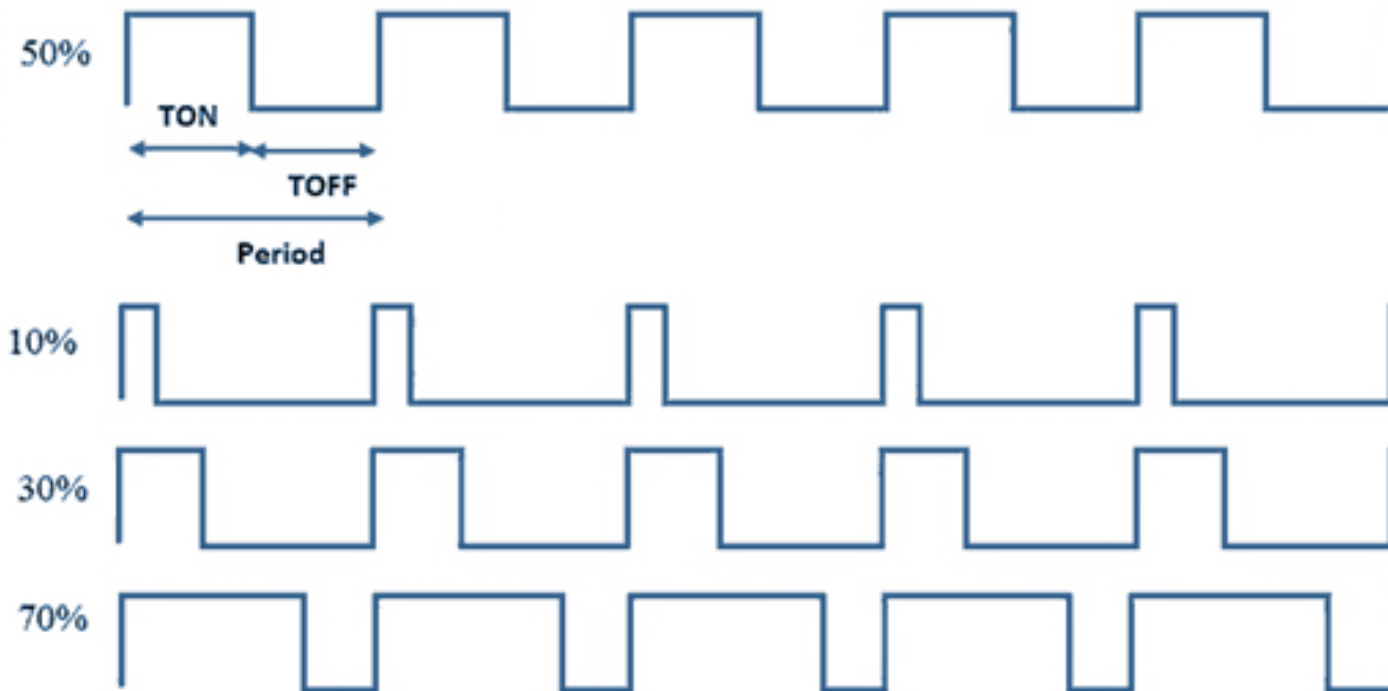
PWM control

To control the speed of the DC motor

- adjust the voltage level applied to its terminal
 - speed is proportional to the voltage applied

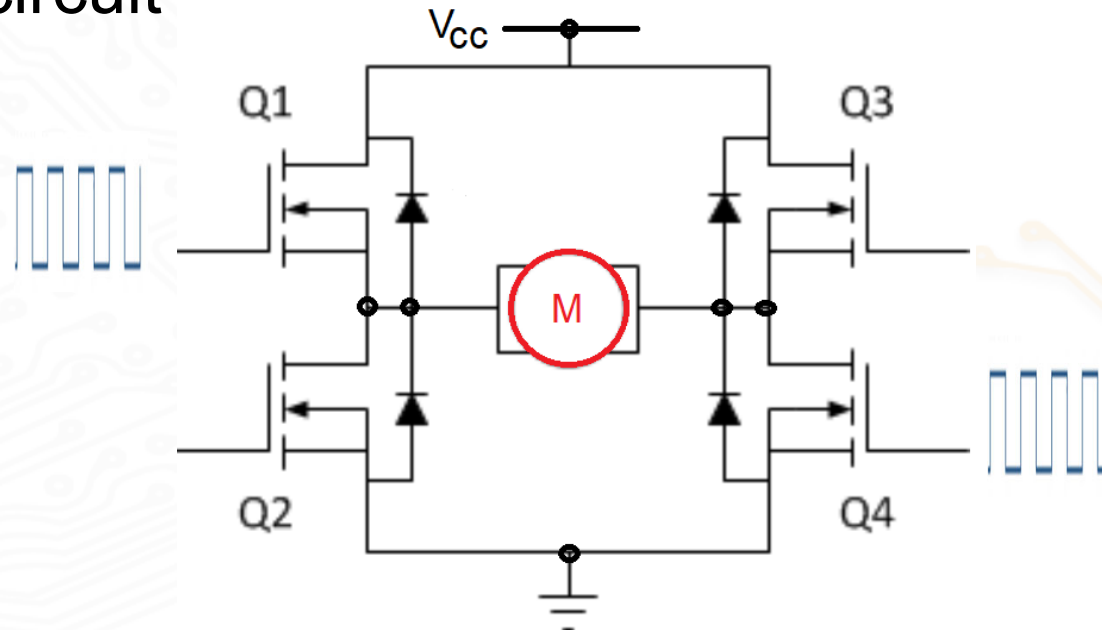
For microcontroller based control system

- Pulse Width Modulation (PWM) control is used
 - average voltage level equals to the PWM duty cycle



H-Bridge Circuit

The PWM waveform is applied to the motor through a H-Bridge circuit



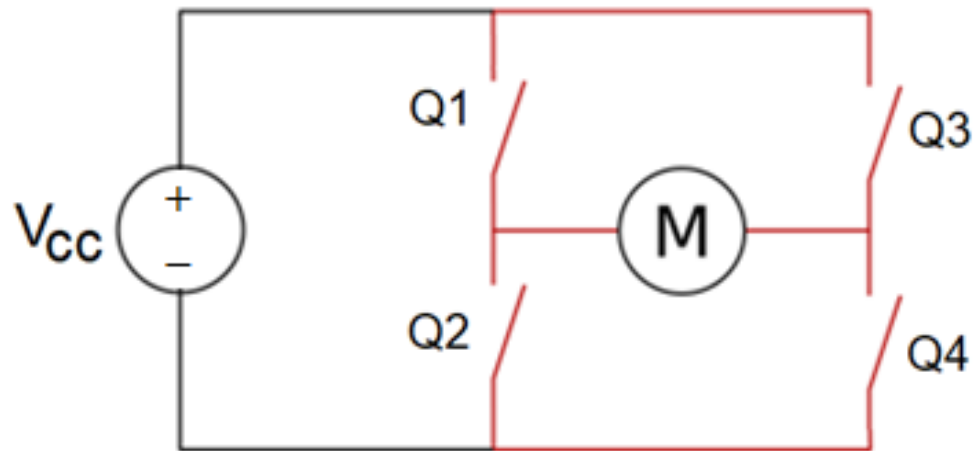
The H-Bridge circuit is commonly available in IC chip for low power DC motor drive

- e.g., AT8236 on the STM32 (ver. D) circuit board

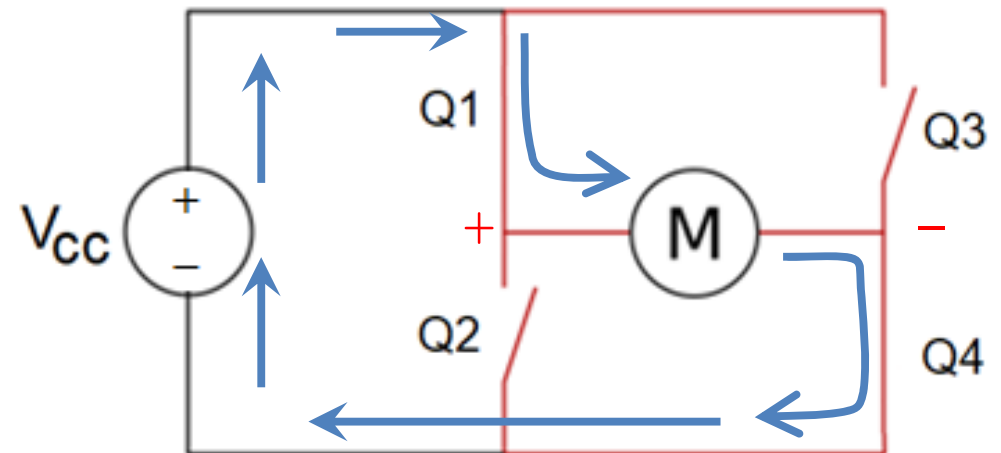


H-Bridge Circuit Operation

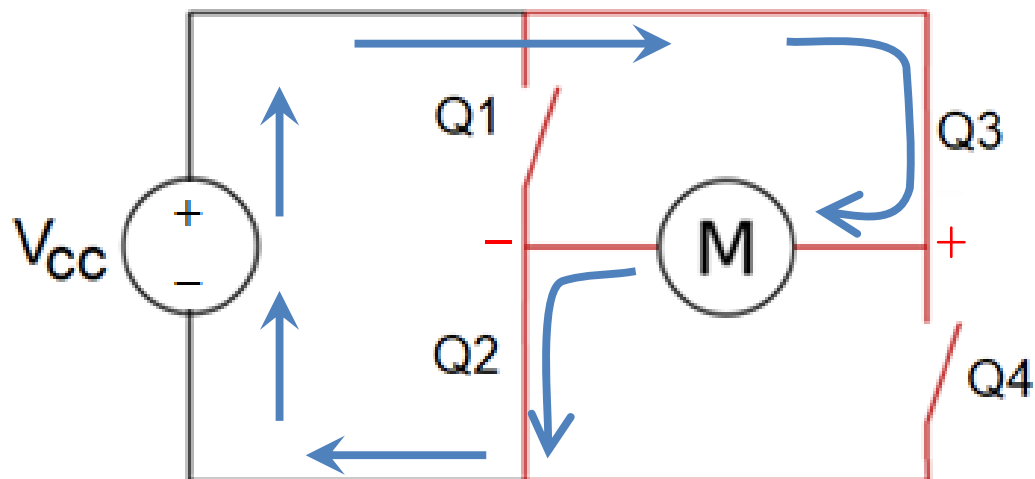
Equivalent circuit



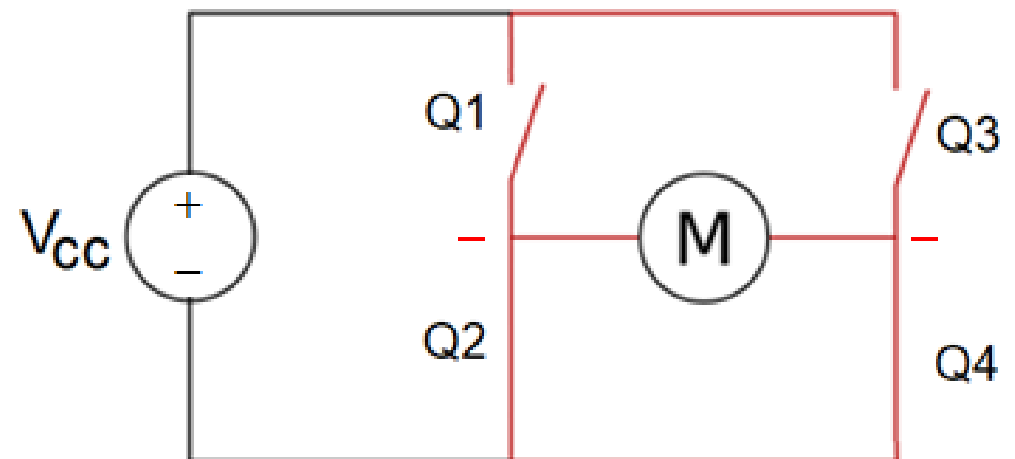
Clockwise rotation



Counter-clockwise rotation



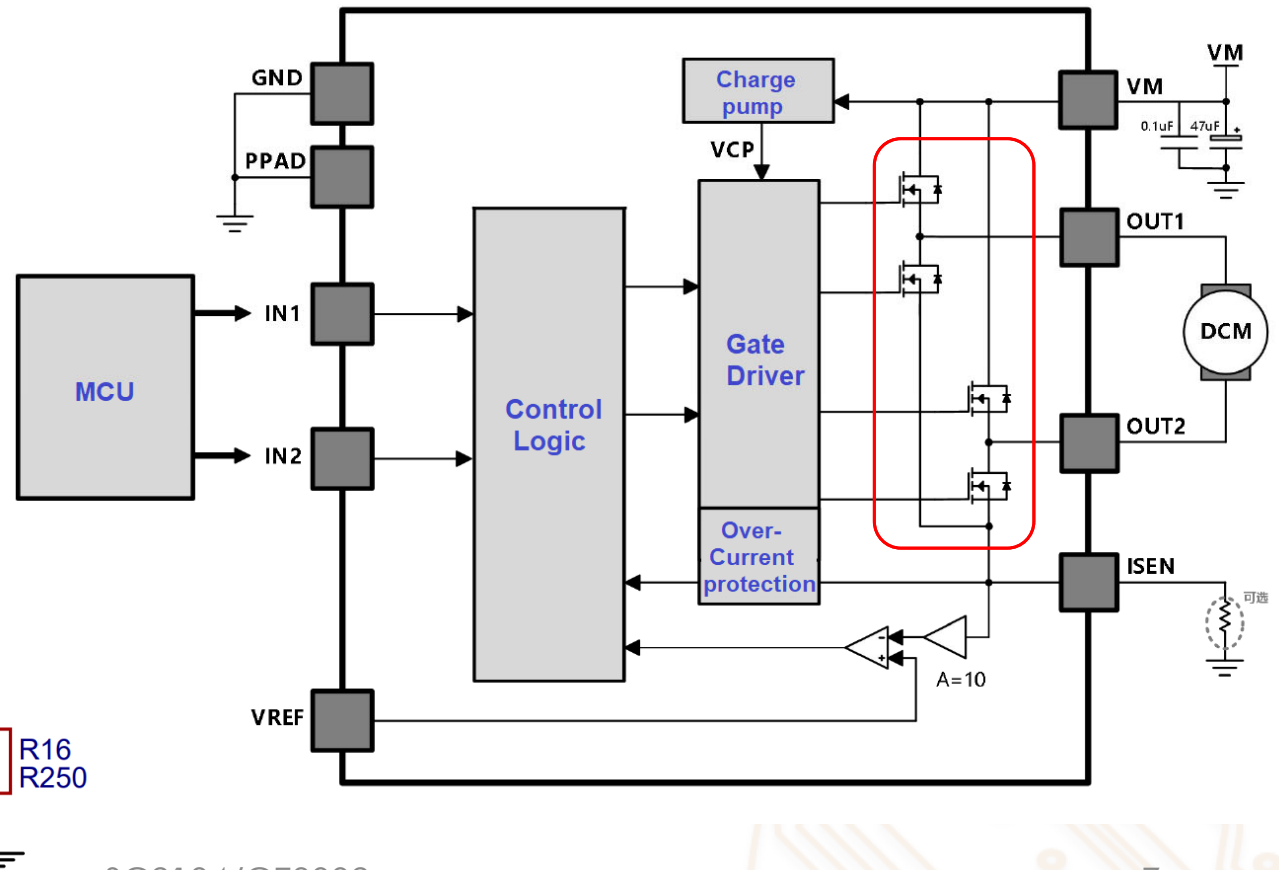
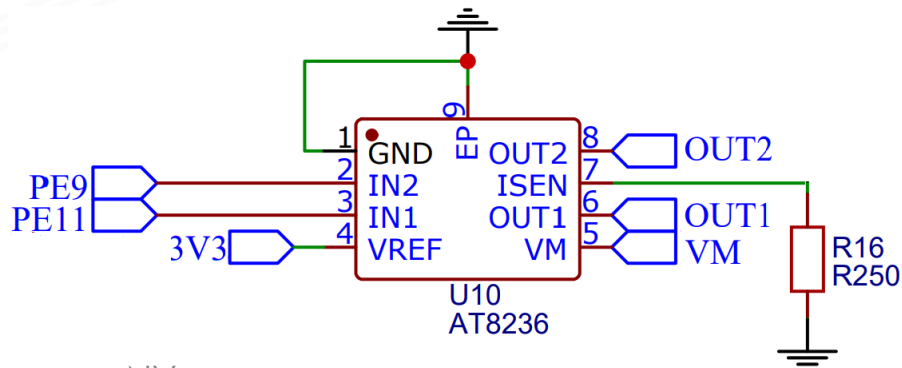
Stop



Case study – AT8236

AT8236 H-Bridge circuit based motor driver IC

- consists of a H-Bridge circuit capable of driving a DC motor
- include the control logic circuits need to control the rotation of the DC motor

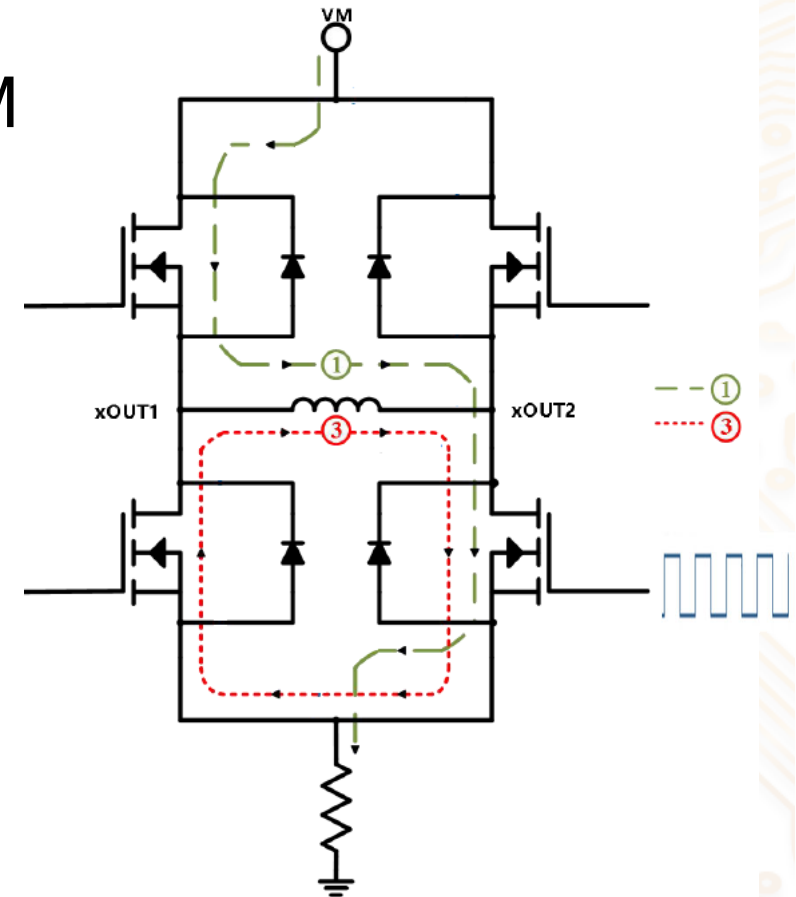
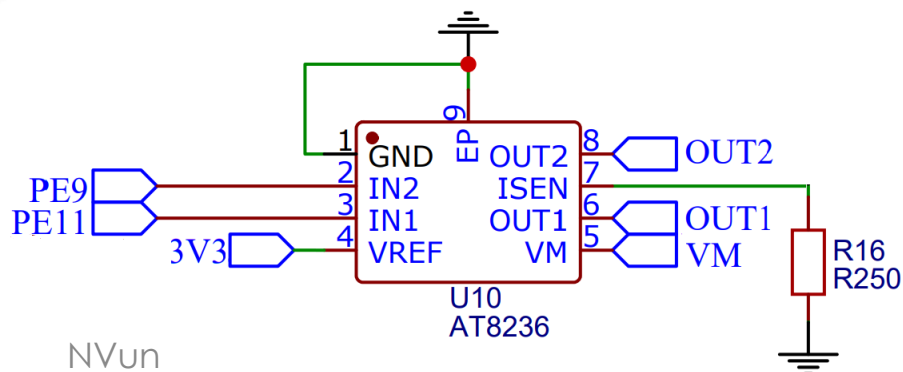


Case study – AT8236 Control Logic

IN1	IN2	OUT1	OUT2	STATE
0	0	Z	Z	Free-Wheeling
1	0	H	L	Forward
0	1	L	H	Reverse
1	1	L	L	Stop

Speed and rotation control using PWM

IN1	IN2	PWM Control
1	PWM	Forward/Clockwise
PWM	1	Reverse/Counter Clockwise
1	1	Brake/Stop



DC Gear Motor

A DC gear motor is an all-in-one combination of a DC motor and a gearbox



Addition of a gearbox to a motor

- reduces its output shaft speed
 - E.g., a 30:1 ratio means that the output shaft rotation is 30 times slower than the DC motor rotor's rotation
- but increase the torque output
 - can move heavier load

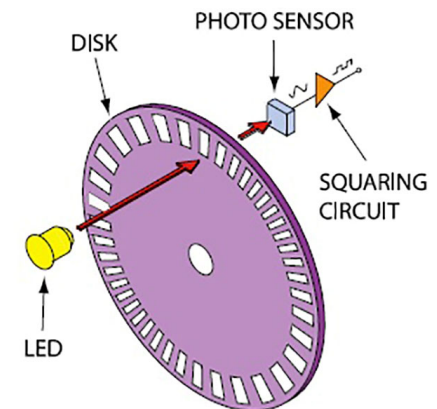
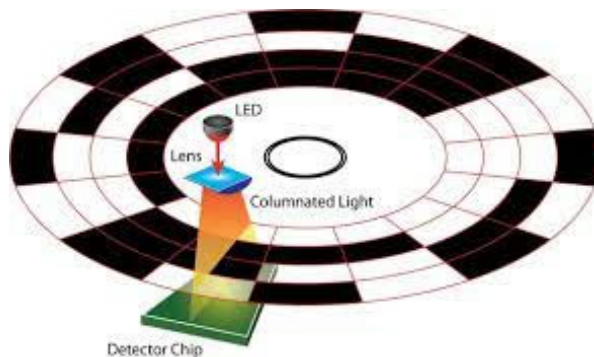
Motor Encoder

In many applications

- it is important to be able to measure the output shaft movement of the motor
 - to derive its position and direction of rotation

Rotary encoders are used for such purpose

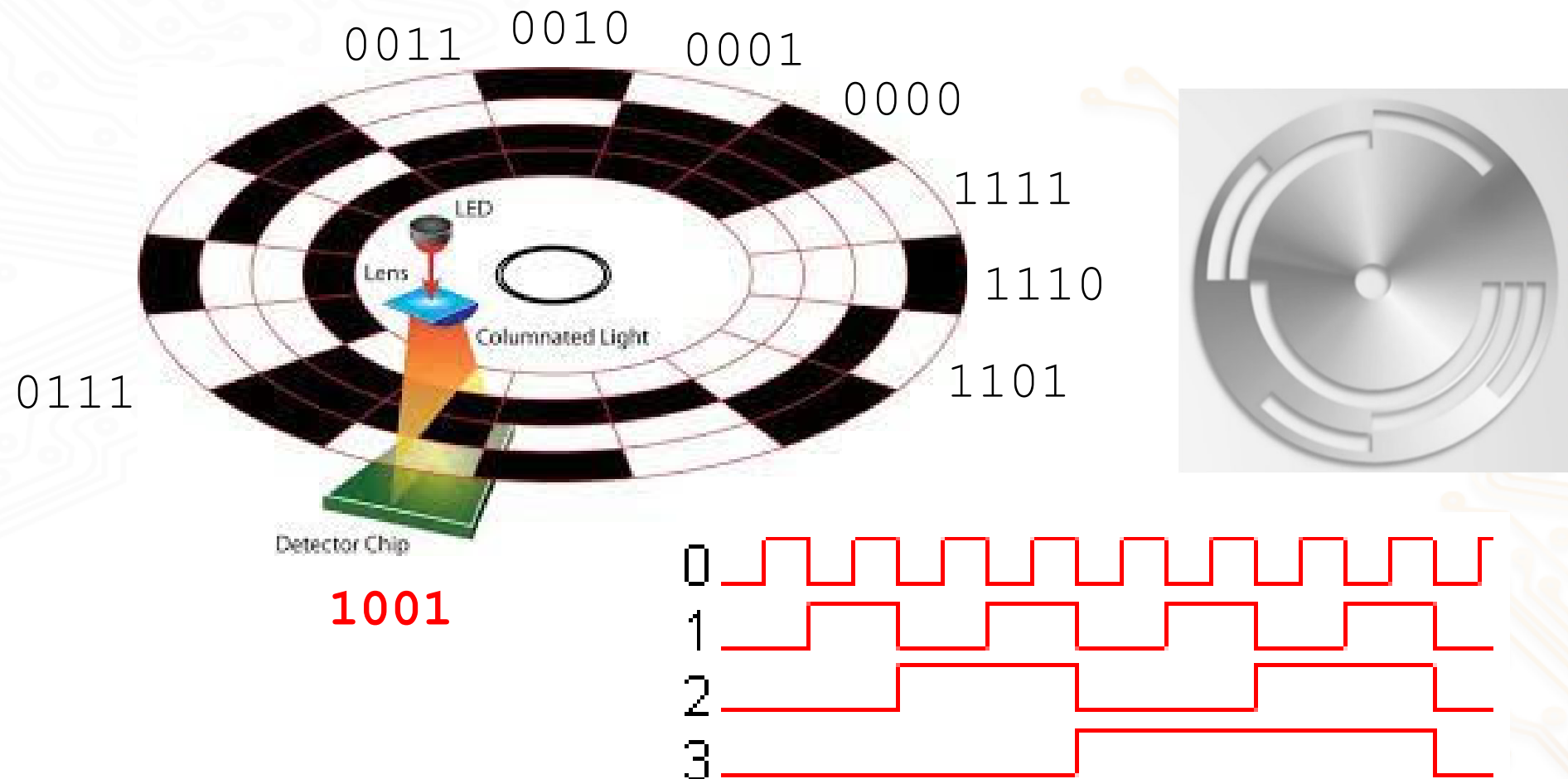
- there are two types of shaft encoder
 - Absolute encoder
 - Incremental encoder
- can be connected to the DC motor shaft or the gearbox output shaft



Absolute Encoder

Absolute encoder can keep track of absolute position of the shaft position

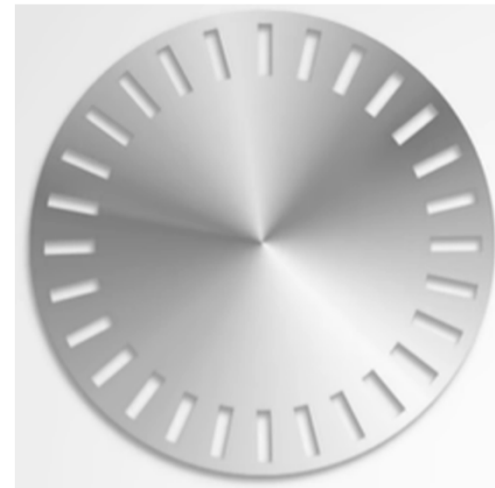
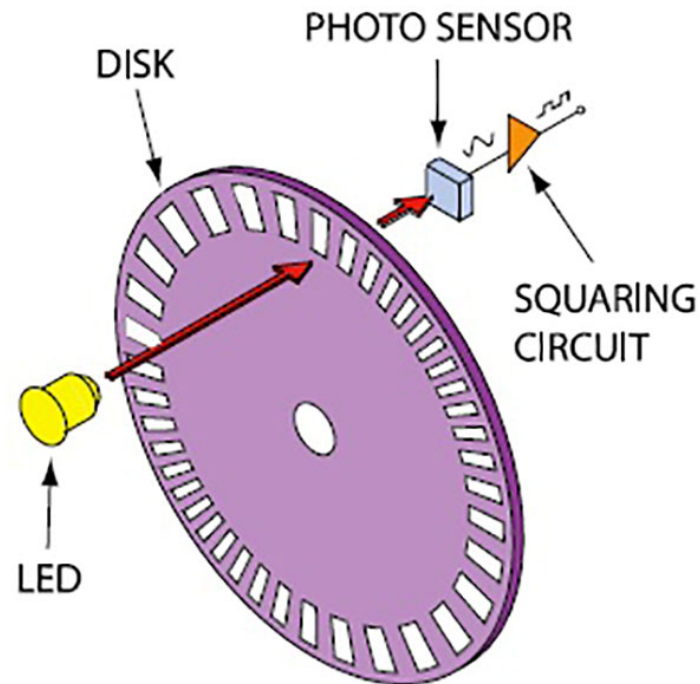
- by assigning a unique digital value to each position



Incremental Encoder

An incremental encoder measures changes in position

- by producing a specific number of equally spaced pulses per revolution



Hall Effect Sensor

Hall effect sensor is a type of solid state based magnetic sensor

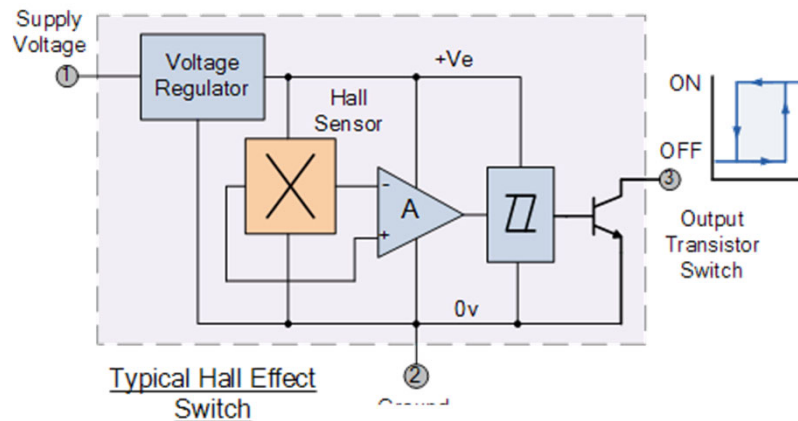
- which can detect the strength and direction of a magnetic field produced from a permanent magnet
- immune to vibration, dust and water



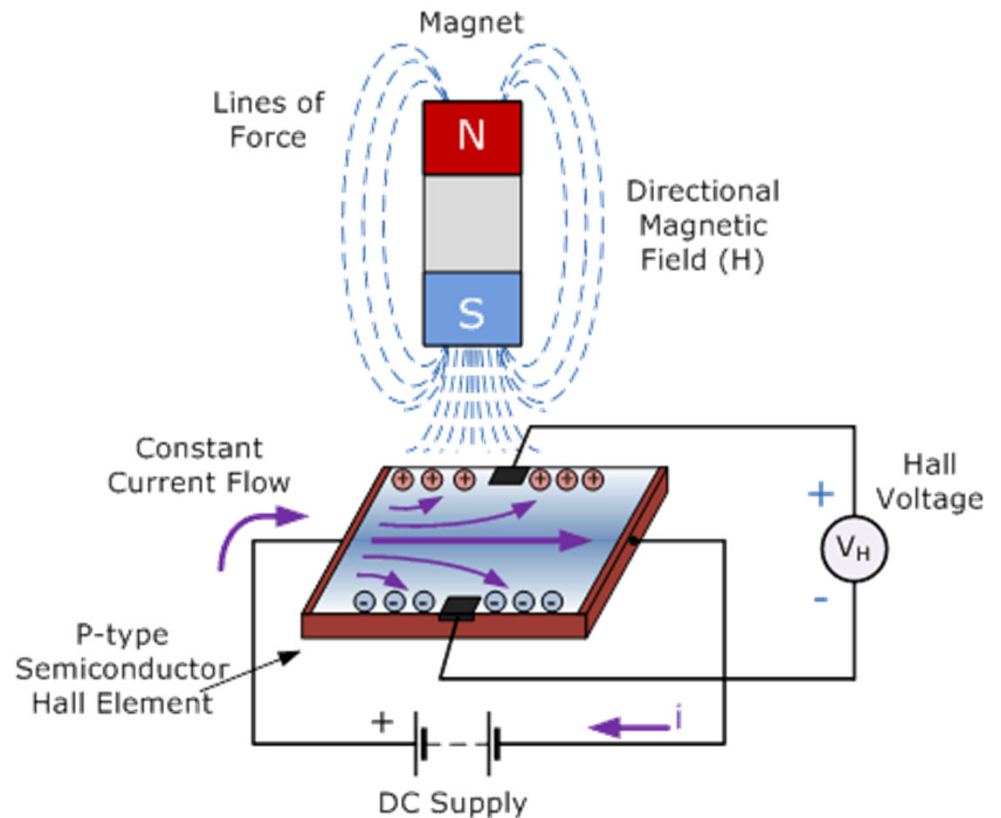
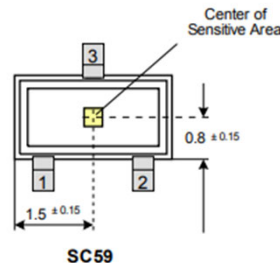
Hall Effect and Hall Voltage

When a Hall element is placed within a magnetic field

- the magnetic flux lines exert a force on the semiconductor material
- deflects the charge carriers (electrons and holes) to either edges of the semiconductor slab
- produce a (Hall) voltage across the two edges



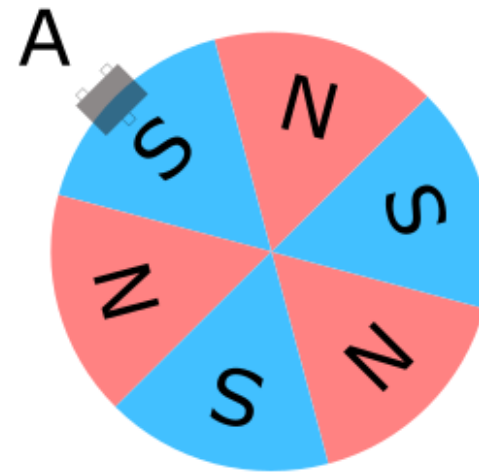
Typical Hall Effect Switch



Hall Effect Rotary Encoder

A hall effect sensor is placed on the PCB

- a magnetic disc is placed on the motor shaft



As the motor shaft and the disc rotate

- pulses are output when the North and South poles pass by the sensor

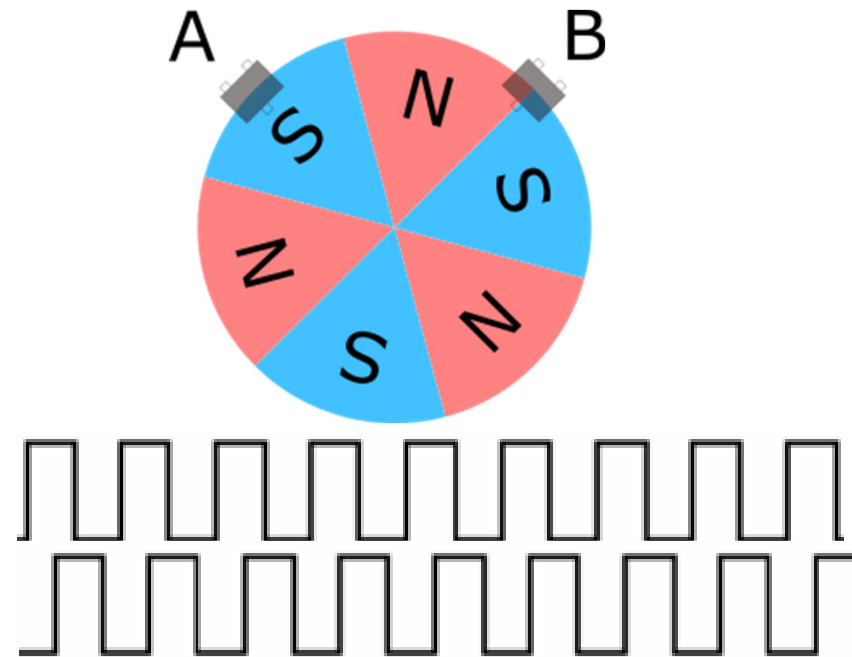
Incremental Encoder with quadrature output

With one output

- an incremental encoder can only determine position

To be able to sense the direction of rotation

- Quadrature encoder uses another sensor
 - outputs a signal that is 90 degree out of phase to the other signal (known as Channels A and B)

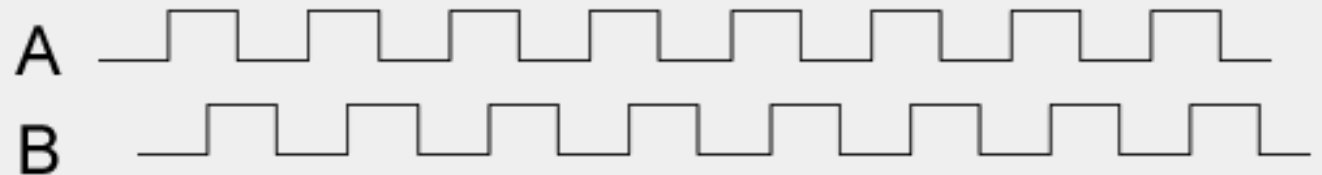


Incremental Encoder with quadrature output

Depending on the direction of the rotation

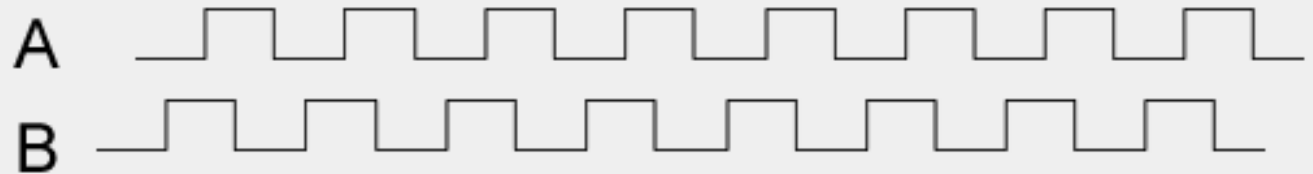
- one channel will be leading the other by 90 degree

Clockwise:

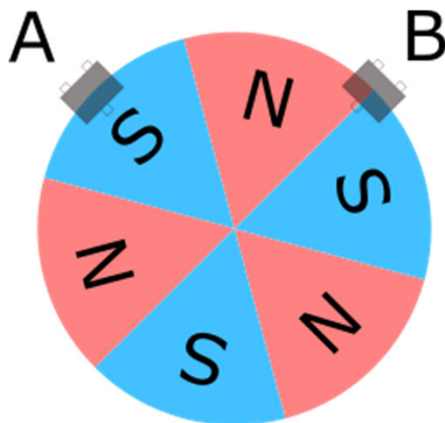


A leads B

Anti-Clockwise:



B leads A



By examining the sequence of channels A and B signals

- can work out the direction and how fast the motor is turning.

Incremental Encoder with quadrature output

By counting the number of edges (e.g. using interrupt)

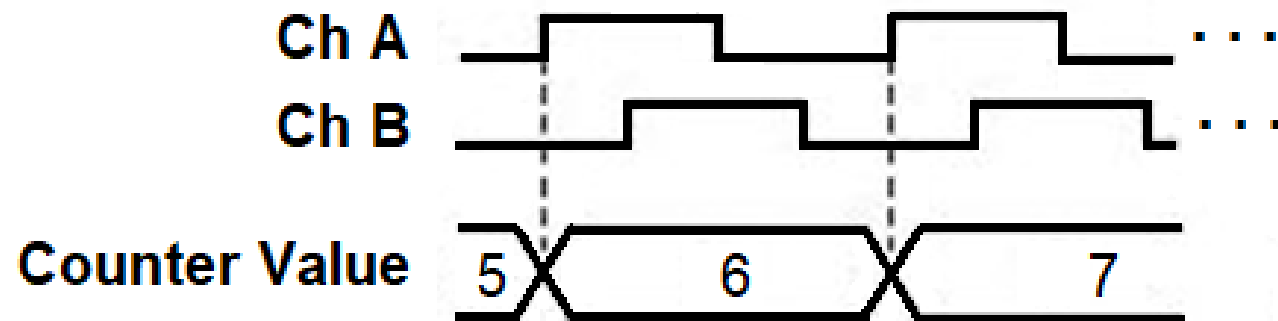
- we can hence determine the motor direction, its speed of rotation and its angular position.

With quadrature output

- three types of encoding, X1, X2 or X4 can be used to achieve different resolution

a) X1 encoding

Only the rising edge (or falling edge) of channel A is counted

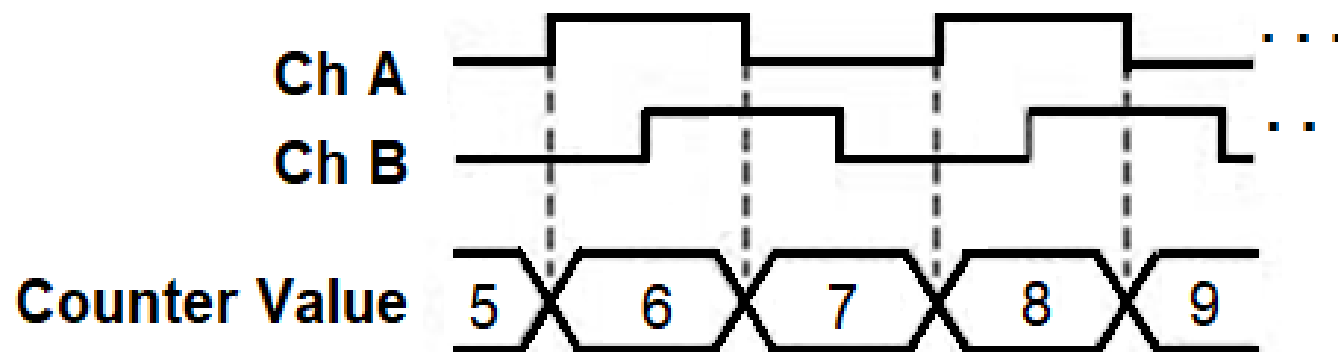


Incremental Encoder with X2 encoding

b) X2 encoding

Both the rising edge and falling edge of channel A are counted

- this doubles the number of pulses counted per revolution
- i.e. the encoder resolution is doubled
- we can determine more precisely the position of the rotating shaft

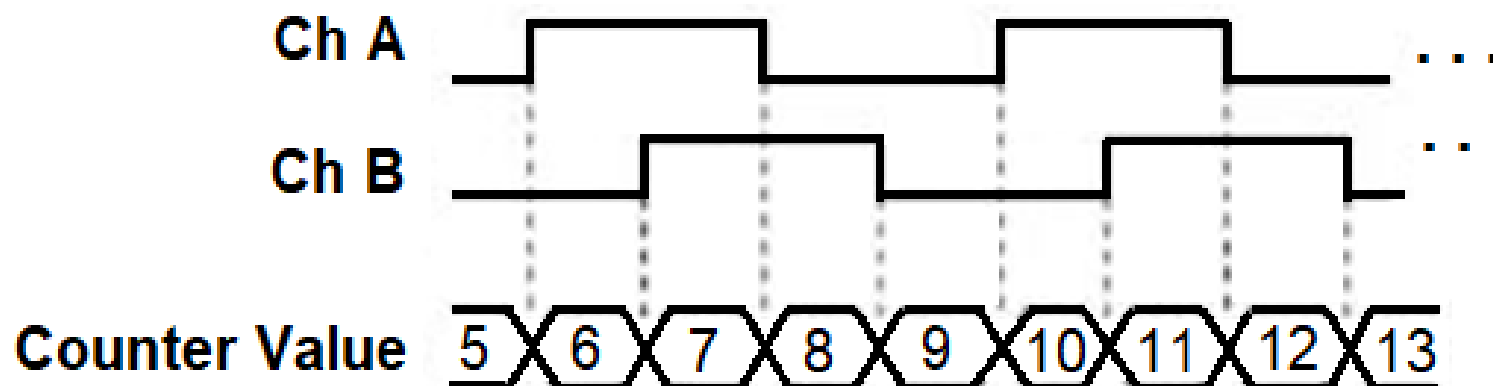


Incremental Encoder with X4 encoding

c) X4 encoding

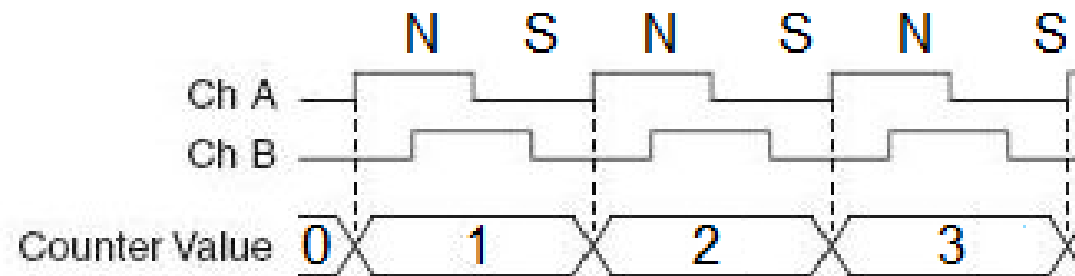
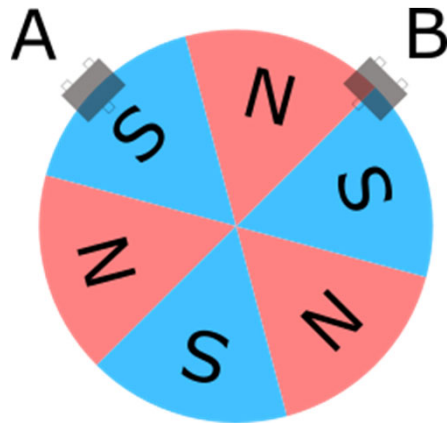
Both the rising edge and falling edge of channel A and channel B are counted

- this doubles the number of pulses counted per revolution
- i.e. the encoder resolution is increased by four times



Incremental Encoder with quadrature output

Knowledge check: A six poles magnet disc



- a) How many pulses are generated per revolution in Channel A?
- b) How many counts are there in one revolution for
 - (i) X1 encoding
 - (ii) X2 encoding
 - (iii) X4 encoding

Servo Motor

A DC **servo motor** is a type of motor that can rotate with great precision



The motor consists of an internal control circuit

- that provides feedback information on the current position of the motor shaft
- adjusts the rotation until the position is equal to the command issued by the user

Enable the servo motor to rotate an object to some specific angles

Servo Motor Interfacing

A **servo motor** uses closed loop control system

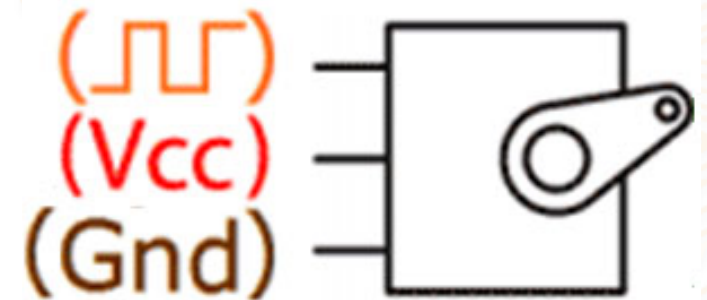
- with position feedback to control the angular position of its output shaft

It is interfaced through 3 wires:

- Orange/Yellow = a PWM signal
- Red = Vcc
- Black/Brown = Ground

The position of the output shaft

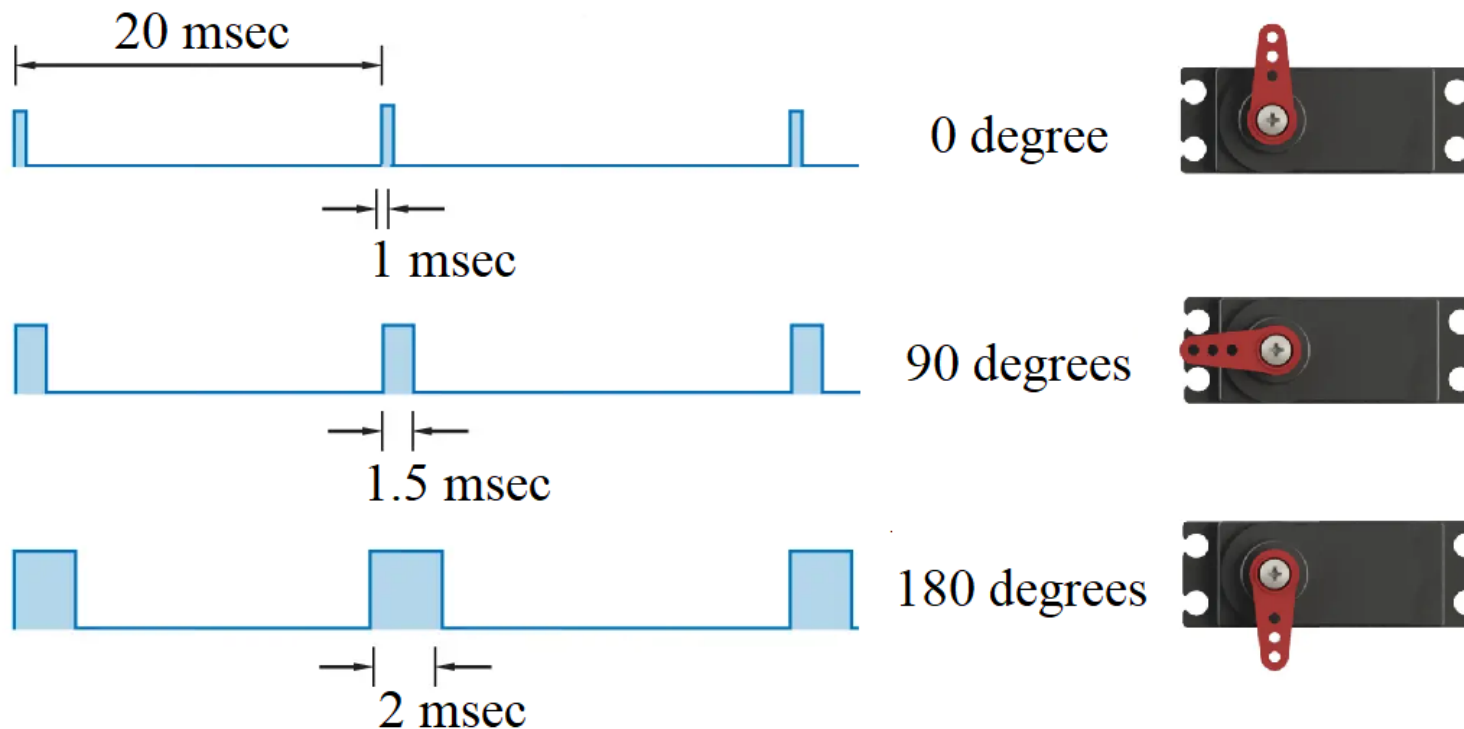
- determined by the PWM signal applied to the orange wire



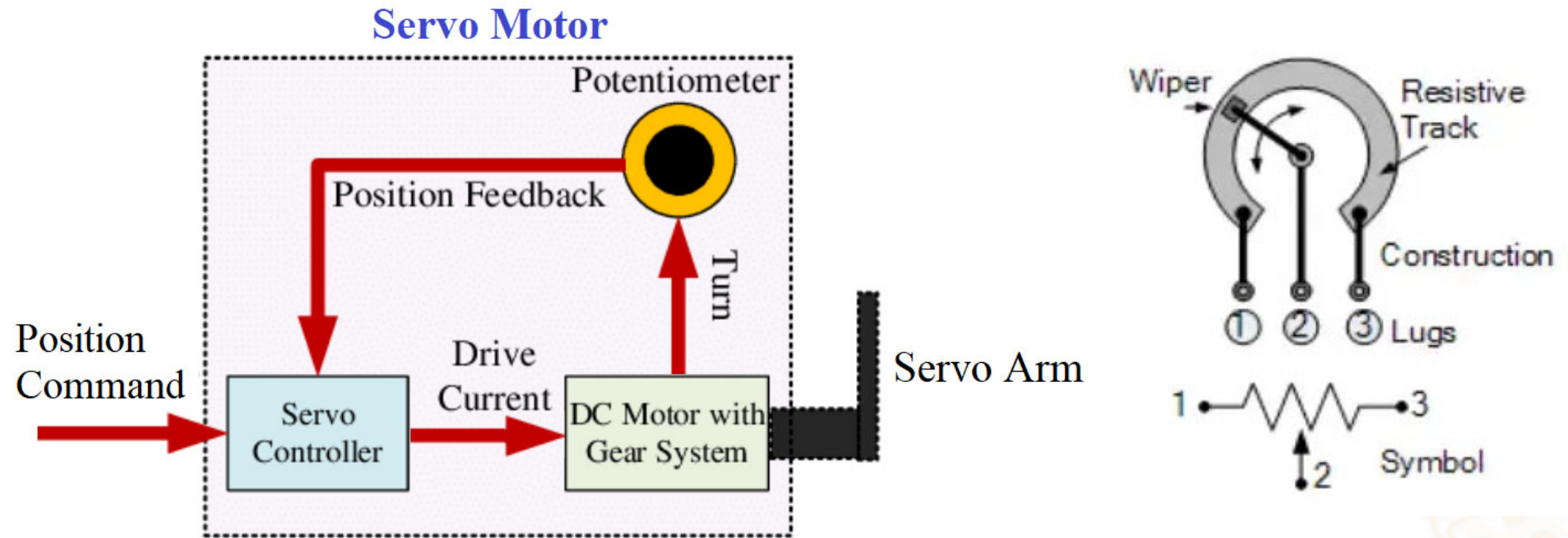
Servo Motor Control

The servo is controlled by the PWM signal as follows

- Period of the PWM signal = 20msec (50Hz)
- Pulse width determines the position of the output shaft
 - 1 msec = 0 degree
 - 1.5 msec = 90 degree
 - 2 msec = 180 degree



Servo Motor Operating Principle



The Servo motor internally uses a potentiometer

- output a voltage that corresponds to the position of the servo arm
- its controller then adjusts the DC motor rotation until the voltage is equal to the **Position Command** issued through the PWM waveform (i.e., its average value)

Summary

Motor's movement is controlled by the voltage applied to its terminals

- can be driven by using PWM waveforms applied through a H-Bridge circuit

Rotary encoder can be used to sense the movement of the motor

- based on optical sensor or Hall effect sensor

Incremental encoder is a relatively simple sensor

- can only measure the position
- direction can be sensed by using Quadrature output
 - resolution can be increased by using X4 encoding

Servo Motor uses internal feedback control loop to achieve very precise angular positioning